Algebra 1 Name Answer Key ___________
Unit 1 Date_____________________
Lesson 4 Homework Period____________________

For questions 1-5, use the exponent laws to write equivalent expressions.

1. $(4.5^2)^5 = 4.5^{10}$

2. $\left(\frac{3}{7}\right)^{-3} = \frac{7^3}{3^3}$

3. $(-2 \cdot 6)^4 = (-2)^4(6)^4$

4. $\frac{(-3.1)^{10}}{(-3.1)^3} = (-3.1)^7$

5. $7^{-4} \cdot 7 \cdot 7^3 = 7^0 = 1$

Find the value of each for questions 6-7.

6. $\sqrt[3]{64} = 64^{\frac{1}{3}} = (4^3)^{\frac{1}{3}} = 4$

7. $\sqrt{64} = 64^{\frac{1}{2}} = (8^2)^{\frac{1}{2}} = 8$

8. Explain why the value of the $\sqrt{64}$ is different from the value of the $\sqrt[3]{64}$.
Different index → Different exponents

For questions 9-10, find the value of the variable that makes the equation true.

9. $\sqrt{y} \cdot \sqrt{y} = 15$

$\sqrt{y^2} = 15$

$y = 15$

$y =$ __15____

10. $\sqrt[3]{g} \cdot \sqrt[3]{g} \cdot \sqrt[3]{g} = -3$

$\sqrt[3]{g^3} = -3$

$g =$ __-3____